

Quiz 1. review.

Proof. $\sqrt{3}$ is irrational

We will need the following lemma.

① Lemma: Any rational $x: \frac{m}{n}$ can be written as a ratio of two integers m, n without common factors

② Lemma: If the square of an integer is divisible by 3, then the integer itself is divisible by 3

Define. if a is an integer and

$$a^2 \equiv 0 \pmod{3}$$

$$\text{if } a \equiv 1, \text{ or } 2 \pmod{3}$$

$$\text{then } a^2 \equiv 1 \pmod{3}$$

$$\text{if } a \equiv 0 \pmod{3}$$

$$\text{then } a^2 \equiv 0 \pmod{3}$$

We assume that $\sqrt{3}$ is a rational

From Lemma ①, we can represent

$\sqrt{3}$ as $\frac{m}{n}$, where m, n

don't have common factors

and $n \neq 0$

We have: $(\sqrt{3})^2 = 3 = \frac{m^2}{n^2}$

$$m^2 = 3n^2$$

From Lemma ①: $3n^2$ is divisible by 3

$\Rightarrow m^2$ is divisible by 3

$\Rightarrow m$ is divisible by 3

Assume that $m = 3k$. $m^2 = 9k^2$

Therefore: $9k^2 = 3n^2 \Rightarrow 3k^2 = n^2$

From Lemma ②: $3k^2$ is divisible by 3

$\Rightarrow n^2$ is divisible by 3

$\Rightarrow n$ is divisible by 3

Therefore, m, n is both divisible by 3

$\Rightarrow m, n$ have a common factor 3

Therefore, we see that $\sqrt{3}$ is not a rational

It is a irrational